MAT 3007 – Optimization

# Duality in LP

*Lecture 11*

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# Recap

# Two-Phase Method in Simplex Tableau

Original problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

Auxiliary problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{y}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0} \end{aligned}$$

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**Phase I:** Solve the auxilary problem with simplex tableau

- $(\boldsymbol{x} = \boldsymbol{0}; \boldsymbol{y} = \boldsymbol{b})$ is an initial BFS for the auxilairy problem with $\boldsymbol{y}$ being basic variables. But it's not in cononical form.

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**Phase I:** Solve the auxilary problem with simplex tableau

- $(\boldsymbol{x} = \boldsymbol{0}; \boldsymbol{y} = \boldsymbol{b})$ is an initial BFS for the auxilairy problem with $\boldsymbol{y}$ being basic variables. But it's not in cononical form.
- Compute upper-left tableau (reduced costs):
  - For basic variables, the reduced costs are 0.
  - For nonbasic variables, $\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j = -\mathbb{1}^\top \boldsymbol{A}_j$, so the $j$-th reduced cost is the negative of the sum of that column.
- Compute upper-right tableau (negative objective): $-\mathbb{1}^\top \boldsymbol{b}$

Compute with $c$ in the auxiliary problem

# Two-Phase Method in Simplex Tableau

Assume that Phase I of the simplex method has been completed, yielding an optimal solution $(\boldsymbol{x}^*, \mathbf{0})$ with an optimal value of 0.

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**Phase II:** Solve the original problem from $\boldsymbol{x}^*$

1. Drop all columns associated with the auxiliary variables.
2. Change the upper-left to the reduced costs of the original problem: $c$ in the original problem

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j$$

already computed in the lower-left tableau

   and the upper-right to the current (negative) objective value.
3. Continue the simplex tableau to solve the original problem.

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   and the upper-right to the current (negative) objective value.
3. Continue the simplex tableau to solve the original problem.

If the output $\boldsymbol{y}^* = \mathbf{0}$ includes basic variables, substitute the basic indices in $\boldsymbol{y}^*$ with some indices of the original variables such that they can form a full basis. (How to do that with Tableau?)

# What If $y^*$ Include Basic Variables

If the output $\boldsymbol{y}^* = \mathbf{0}$ after Phase I includes basic variables, we can substitute them with some indices of the original variables :

**How to do it with a tableau?** After Phase I, assume $B(\ell)$ is in $\boldsymbol{y}^*$. We do the following to the tableau from Phase I.

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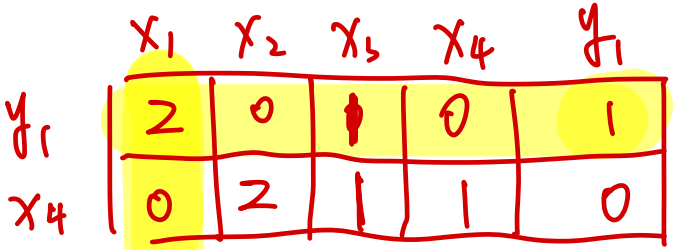
- We ignore the upper section of the tableau.
- Find a column in $\boldsymbol{x}$ such that $\ell$-th element is not 0. Select it as the pivot column (make it enter the basis) and swap the current $B(\ell)$ out by doing row operations. Note that this does not change the current BFS.

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- We ignore the upper section of the tableau.
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- Repeat until we swap all basis in $\boldsymbol{y}^*$ out. Then we move to Phase II.



# Example

$$
\begin{array}{lrrrrcl}
\text{minimize} & x_1 & +x_2 & +x_3 & & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & = & 3 \\
 & & -4x_2 & -9x_3 & & = & -5 \\
 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1, & & x_2, & x_3, \quad x_4 & \geq & 0
\end{array}
$$

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 & & -4x_2 & -9x_3 & & = & -5 \\
 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1, & x_2, & x_3, & x_4 & \geq & 0
\end{array}
$$

First, make $\boldsymbol{b}$ positive and construct the auxiliary problem:

$$
\begin{array}{lrrrrrrrcl}
\text{minimize} & & & & & x_5 & +x_6 & +x_7 & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & +x_5 & & & = & 3 \\
 & & 4x_2 & +9x_3 & & & +x_6 & & = & 5 \\
 & & & +3x_3 & +x_4 & & & +x_7 & = & 1 \\
 & x_1, & x_2, & x_3, & x_4, & x_5, & x_6, & x_7 & \geq & 0
\end{array}
$$

# Example Continued

Construct the initial tableau for the auxiliary problem

| B | -1 | -6 | -15 | -1 | 0 | 0 | 0 | -9 |
|---|---|---|---|---|---|---|---|---|
| 5 | 1 | 2 | 3 | 0 | 1 | 0 | 0 | 3 |
| 6 | 0 | 4 | 9 | 0 | 0 | 1 | 0 | 5 |
| 7 | 0 | 0 | 3 | 1 | 0 | 0 | 1 | 1 |

# Example Continued

Construct the initial tableau for the auxiliary problem

| B | -1 | -6 | -15 | -1 | 0 | 0 | 0 | -9 |
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| 5 | 1 | 2 | 3 | 0 | 1 | 0 | 0 | 3 |
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| 7 | 0 | 0 | 3 | 1 | 0 | 0 | 1 | 1 |

Carry out the simplex method: Iteration 1

| B | 0 | -4 | -12 | -1 | 1 | 0 | 0 | -6 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 3 | 0 | 1 | 0 | 0 | 3 |
| 6 | 0 | 4 | 9 | 0 | 0 | 1 | 0 | 5 |
| 7 | 0 | 0 | 3 | 1 | 0 | 0 | 1 | 1 |

# Example Continued

Iteration 2:

| B | 0 | 0 | -3 | -1 | 1 | 1 | 0 | -1 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | -3/2 | 0 | 1 | -1/2 | 0 | 1/2 |
| 2 | 0 | 1 | 9/4 | 0 | 0 | 1/4 | 0 | 5/4 |
| 7 | 0 | 0 | 3 | 1 | 0 | 0 | 1 | 1 |

# Example Continued

Iteration 2:

| B | 0 | 0 | -3 | -1 | 1 | 1 | 0 | -1 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | -3/2 | 0 | 1 | -1/2 | 0 | 1/2 |
| 2 | 0 | 1 | 9/4 | 0 | 0 | 1/4 | 0 | 5/4 |
| 7 | 0 | 0 | 3 | 1 | 0 | 0 | 1 | 1 |

Iteration 3:

| B | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1/2 | 1 | -1/2 | 1/2 | 1 |
| 2 | 0 | 1 | 0 | -3/4 | 0 | 1/4 | -3/4 | 1/2 |
| 3 | 0 | 0 | 1 | 1/3 | 0 | 0 | 1/3 | 1/3 |

This is optimal for the auxiliary problem. $\boldsymbol{x} = (1; \frac{1}{2}; \frac{1}{3}; 0)$ is a BFS for the original problem ($B = \{1, 2, 3\}$).

# Example Continued

| B | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1/2 | 1 | -1/2 | 1/2 | 1 |
| 2 | 0 | 1 | 0 | -3/4 | 0 | 1/4 | -3/4 | 1/2 |
| 3 | 0 | 0 | 1 | 1/3 | 0 | 0 | 1/3 | 1/3 |

Drop all the columns for auxiliary variables. We recompute the reduced cost for the original problem:

$$\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A} = (0, 0, 0, -\tfrac{1}{12}).$$

and the objective value: $\frac{11}{6}$.

$\bar{c}_4 = c_4 - c_B^\top A_B^{-1} A_4$

$= 0 - (1\ 1\ 1)^\top \begin{pmatrix} \frac{1}{2} \\ -\frac{3}{4} \\ 1/3 \end{pmatrix}$

$= -\frac{1}{12}$

basic

Now the simplex tableau becomes:

| B | 0 | 0 | 0 | -1/12 | -11/6 |
|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1/2 | 1 |
| 2 | 0 | 1 | 0 | -3/4 | 1/2 |
| 3 | 0 | 0 | 1 | 1/3 | 1/3 |

# Example Continued

Then we continue from the new simplex tableau:

| B | 0 | 0 | 0 | -1/12 | -11/6 |
|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1/2 | 1 |
| 2 | 0 | 1 | 0 | -3/4 | 1/2 |
| 3 | 0 | 0 | 1 | 1/3 | 1/3 |

# Example Continued

Then we continue from the new simplex tableau:

| B | 0 | 0 | 0 | -1/12 | -11/6 |
|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1/2 | 1 |
| 2 | 0 | 1 | 0 | -3/4 | 1/2 |
| 3 | 0 | 0 | 1 | 1/3 | 1/3 |

The next pivot:

| B | 0 | 0 | 1/4 | 0 | -7/4 |
|---|---|---|---|---|---|
| 1 | 1 | 0 | -3/2 | 0 | 1/2 |
| 2 | 0 | 1 | 9/4 | 0 | 5/4 |
| 4 | 0 | 0 | 3 | 1 | 1 |

This is optimal and we can stop. The optimal solution is $\boldsymbol{x} = (\frac{1}{2}; \frac{5}{4}; 0; 1)$. The optimal value is $\frac{7}{4}$.

# Duality Theory for LPs

# Duality Theory: Motivation

We consider the standard LP ($m$ constraints, $n$ variables):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

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We can write it as:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} + \max_{\boldsymbol{y} \in \mathbb{R}^m} \boldsymbol{y}^\top (\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

$\sum_i y_i(b_i - a_i^\top x)$

if $b_i > a_i^\top x$, we let $y_i \to \infty$

if $b_i < a_i^\top x$, we let $y_i \to -\infty$

**Why?**

- If $\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$, then we can let $\boldsymbol{y}^\top (\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) \to \infty$ by choosing $\boldsymbol{y}$, and such $\boldsymbol{x}$ cannot be optimal. Hence, it implicitly enforces the constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. If $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, then $\boldsymbol{y}^\top (\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}) = 0$, so they attain the same objective value.

$$\min_{\boldsymbol{x}\geq \mathbf{0}} \quad \max_{\boldsymbol{y}} \quad \boldsymbol{c}^\top \boldsymbol{x} + \boldsymbol{y}^\top (\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x})$$

$\rightsquigarrow$ Now we assume that we can exchange max and min (we will justify it later). Then the problem becomes:

$$\max_{\boldsymbol{y}} \quad \boldsymbol{b}^\top \boldsymbol{y} + \min_{\boldsymbol{x}\geq \mathbf{0}} \; \boldsymbol{x}^\top (\boldsymbol{c} - \boldsymbol{A}^\top \boldsymbol{y})$$

We claim that this is equivalent to:

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}. \end{aligned}$$

**Why?**

$$\min_{\boldsymbol{x}\geq \mathbf{0}} \; \boldsymbol{x}^\top (\boldsymbol{c} - \boldsymbol{A}^\top \boldsymbol{y}) = \begin{cases} 0 & \text{if } \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}, \\ -\infty & \text{if } \boldsymbol{A}^\top \boldsymbol{y} \not\leq \boldsymbol{c}. \end{cases}$$

# Two Linear Programs . . .

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0} \end{aligned}$$

and

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c} \end{aligned}$$

We call them dual to each other. If we call one the primal problem, then the other one is called the dual problem.

- We call $\boldsymbol{y}$ the dual variables (to the first LP).

$\rightsquigarrow$ By our derivation, they should have the same optimal value (provided the exchange of min and max is valid).

# General Dual Problems

| Primal | | | Dual | | |
|---|---|---|---|---|---|
| min | $\boldsymbol{c}^\top \boldsymbol{x}$ | | max | $\boldsymbol{b}^\top \boldsymbol{y}$ | |
| s.t. | $\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i,$ | $i \in M_1,$ | s.t. | $y_i \geq 0,$ | $i \in M_1$ |
| | $\boldsymbol{a}_i^\top \boldsymbol{x} \leq b_i,$ | $i \in M_2,$ | | $y_i \leq 0,$ | $i \in M_2$ |
| | $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i,$ | $i \in M_3,$ | | $y_i$ free, | $i \in M_3$ |
| | $x_j \geq 0,$ | $j \in N_1,$ | | $\boldsymbol{A}_j^\top \boldsymbol{y} \leq c_j,$ | $j \in N_1$ |
| | $x_j \leq 0,$ | $j \in N_2,$ | | $\boldsymbol{A}_j^\top \boldsymbol{y} \geq c_j,$ | $j \in N_2$ |
| | $x_j$ free, | $j \in N_3,$ | | $\boldsymbol{A}_j^\top \boldsymbol{y} = c_j,$ | $j \in N_3$ |

- $\boldsymbol{a}_i^\top$ is the $i$th row of $\boldsymbol{A}$, $\boldsymbol{A}_j$ is the $j$th column of $\boldsymbol{A}$.
- Each primal constraint corresponds to a dual variable.
- Each primal variable corresponds to a dual constraint.
- Equality constraints always correspond to free variables.

# To Memorize: Table & Signs

if your primal problem is max, look at the left first.

| ~~Primal~~ | minimize | maximize | ~~Dual~~ |
|---|---|---|---|
| constraints | $\geq b_i$ | $\geq 0$ | variables |
| | $\leq b_i$ | $\leq 0$ | |
| | $= b_i$ | free | |
| variables | $\geq 0$ | $\leq c_j$ | constraints |
| | $\leq 0$ | $\geq c_j$ | |
| | free | $= c_j$ | |

# Example

Consider the following problem:

$$
\begin{array}{lrrrl}
\text{minimize} & x_1 & +2x_2 & +3x_3 & \\
\text{subject to} & -x_1 & +3x_2 & & \leq 5 \\
& 2x_1 & -x_2 & +3x_3 & \geq 6 \\
& & & x_3 & \leq 4 \\
& x_1 \geq 0 & x_2 \leq 0 & x_3 \text{ free} &
\end{array}
$$

The dual is:

$$
\begin{array}{lrrrl}
\text{maximize} & 5y_1 & +6y_2 & +4y_3 & \\
\text{subject to} & -y_1 & +2y_2 & & \leq 1 \\
& 3y_1 & -y_2 & & \geq 2 \\
& & +3y_2 & +y_3 & = 3 \\
& y_1 \text{ free} & y_2 \geq 0 & y_3 \leq 0 &
\end{array}
$$

# One More Example

Recall the support vector machine problem. The primal problem is:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},y,\boldsymbol{t}} \quad & \textstyle\sum_{i=1}^m t_i \\ \text{subject to} \quad & b_i(\boldsymbol{a}_i^\top \boldsymbol{x} + y) + t_i \geq 1, \quad \forall\, i = 1,\ldots,m \\ & t_i \geq 0 \qquad\qquad\quad \forall\, i = 1,\ldots,m \end{aligned}$$

with variables $\boldsymbol{x} \in \mathbb{R}^n$, $y \in \mathbb{R}$, and $\boldsymbol{t} \in \mathbb{R}^m$.

- $\rightsquigarrow$ Associate $u_i$ to each of the constraints.
- The dual problems has $n + m + 1$ constraints.
- Defining the data matrix $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \ldots & \boldsymbol{a}_m \end{pmatrix} \in \mathbb{R}^{n\times m}$, the primal constraints can be written compactly:

$$\begin{pmatrix} \boldsymbol{A}\,\mathrm{diag}(\boldsymbol{b})\,\boldsymbol{u} \\ \boldsymbol{b}^\top \boldsymbol{u} \\ \boldsymbol{u} \end{pmatrix} \begin{cases} = 0 \\ = 0 \\ \leq 1 \end{cases} \qquad \underbrace{\begin{pmatrix} \mathrm{diag}(\boldsymbol{b})\boldsymbol{A}^\top & \boldsymbol{b} & \boldsymbol{I} \end{pmatrix}}_{\tilde{\boldsymbol{A}}} \begin{pmatrix} \boldsymbol{x} \\ y \\ \boldsymbol{t} \end{pmatrix} \geq \mathbb{1}.$$

$\boldsymbol{u} \geqslant 0$

# Example Continued

The dual constraint becomes

$$\begin{pmatrix} \boldsymbol{A}\,\mathrm{diag}(\boldsymbol{b}) \\ \boldsymbol{b}^\top \\ \boldsymbol{I} \end{pmatrix} \boldsymbol{u} = \begin{pmatrix} \boldsymbol{A}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{u} \\ \boldsymbol{b}^\top \boldsymbol{u} \\ \boldsymbol{u} \end{pmatrix} \begin{array}{lll} \Longrightarrow & = \mathbf{0} & (\boldsymbol{x} \text{ free}) \\ \Longrightarrow & = 0 & (y \text{ free}) \\ \Longrightarrow & \leq \mathbb{1} & (\boldsymbol{t} \geq \mathbf{0}) \end{array}$$

with additional constraints $\boldsymbol{u} \geq \mathbf{0}$.

The dual problem is

$$\begin{array}{lll} \text{maximize}_{\boldsymbol{u}} & \sum_{i=1}^m u_i & \\ \text{subject to} & \boldsymbol{A}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{u} = \mathbf{0} & \\ & \boldsymbol{b}^\top \boldsymbol{u} = 0 & \\ & 0 \leq u_i \leq 1, & \forall\ i = 1, \ldots, m \end{array}$$

# Example: Summary

**SVM — Primal Problem:**

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},y,\boldsymbol{t}} \quad & \mathbb{1}^\top \boldsymbol{t} \\ \text{subject to} \quad & \operatorname{diag}(\boldsymbol{b})\boldsymbol{A}^\top \boldsymbol{x} + \boldsymbol{b}y + \boldsymbol{t} \geq \mathbb{1} \\ & \boldsymbol{t} \geq \boldsymbol{0} \end{aligned}$$

**SVM — Dual Problem:**

$$\begin{aligned} \text{maximize}_{\boldsymbol{u}} \quad & \mathbb{1}^\top \boldsymbol{u} \\ \text{subject to} \quad & \boldsymbol{A}\operatorname{diag}(\boldsymbol{b})\boldsymbol{u} = \boldsymbol{0} \\ & \boldsymbol{b}^\top \boldsymbol{u} = 0 \\ & \boldsymbol{u} \geq \boldsymbol{0},\ \boldsymbol{u} \leq \mathbb{1} \end{aligned}$$

- $\boldsymbol{A}$ is the data matrix, $\boldsymbol{b}$ are the labels ($\pm 1$).

# Duality Theory

# Invariance of Transformations

## Theorem: Duality under Transformations

If we transform a linear program to an equivalent one (e.g., by replacing free variables, adding slack variables, etc.), then the dual of the two problems will be equivalent.

## Theorem: Dual$^2$

If we transform the primal to its dual and transform the resulting dual to its dual, then we will obtain a problem equivalent to the primal problem, i.e., the dual of dual is the primal.

# Duality Theorems

By construction, the primal and the dual problem should have the same optimal value.

- In the following, we will formally prove this claim.
- Due to the transformation invariance, we only need to discuss the standard form. The next theorems also apply to general cases as well.

# Weak Duality Theorem

| Primal | | Dual | |
|---|---|---|---|
| min | $\boldsymbol{c}^\top \boldsymbol{x}$ | max | $\boldsymbol{b}^\top \boldsymbol{y}$ |
| s.t. | $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}$ | s.t. | $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$ |

## Weak Duality Theorem

If $\boldsymbol{x}$ is feasible for the primal problem and $\boldsymbol{y}$ is feasible for the dual, then we have:

$$\boldsymbol{b}^\top \boldsymbol{y} \leq \boldsymbol{c}^\top \boldsymbol{x}.$$

# Weak Duality Theorem

| Primal | | Dual | |
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| min | $\boldsymbol{c}^\top \boldsymbol{x}$ | max | $\boldsymbol{b}^\top \boldsymbol{y}$ |
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## Weak Duality Theorem

If $\boldsymbol{x}$ is feasible for the primal problem and $\boldsymbol{y}$ is feasible for the dual, then we have:

$$\boldsymbol{b}^\top \boldsymbol{y} \leq \boldsymbol{c}^\top \boldsymbol{x}.$$

W.L.O.G, we consider (primal $\rightsquigarrow$ min, dual $\rightsquigarrow$ max):

- Any dual feasible solution will give a lower bound on the primal optimal value.
- Any primal feasible solution will give an upper bound on the dual optimal value.
- The optimal value of the primal is larger than that of the dual.

**Proof:**

Assume $\boldsymbol{x}$ is a feasible point of the primal problem and let $\boldsymbol{y}$ be a feasible dual variable. Then we have:

$$\boldsymbol{b}^\top \boldsymbol{y} = (\boldsymbol{A}\boldsymbol{x})^\top \boldsymbol{y} = \boldsymbol{x}^\top (\boldsymbol{A}^\top \boldsymbol{y}) \leq \boldsymbol{c}^\top \boldsymbol{x}.$$

feasibility of primal

$\geq 0$ $\quad$ $\leq \boldsymbol{c}$

The last inequality follows from $\boldsymbol{x} \geq \boldsymbol{0}$ and $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$. $\square$

**Proof:**

Assume $\boldsymbol{x}$ is a feasible point of the primal problem and let $\boldsymbol{y}$ be a feasible dual variable. Then we have:

$$\boldsymbol{b}^\top \boldsymbol{y} = (\boldsymbol{A}\boldsymbol{x})^\top \boldsymbol{y} = \boldsymbol{x}^\top (\boldsymbol{A}^\top \boldsymbol{y}) \leq \boldsymbol{c}^\top \boldsymbol{x}.$$

The last inequality follows from $\boldsymbol{x} \geq \boldsymbol{0}$ and $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$. □

if dual is feasible, say $y$ is feasible, then $A^T y$ is a lower bound for primal → contradiction.

**Corollary: Feasibility and Boundedness**

- If the primal problem is unbounded (i.e., the optimal value is $-\infty$), then the dual problem must be infeasible.
- If the dual problem is unbounded (i.e., the optimal value is $+\infty$), then the primal problem must be infeasible.

## Corollary: Optimality and Duality

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problem, respectively. If $\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{b}^\top \boldsymbol{y}$, then $\boldsymbol{x}$ and $\boldsymbol{y}$ must be optimal solutions of the primal and dual, respectively.

Denote them as $x^*$, and $y^*$.

$$c^T x \geq b^T y^* = c^T x^*$$

$\forall$ feasible $x$ $\Rightarrow c^T x^* \leq c^T x$

$\Rightarrow x^*$ is optimal

## Corollary: Optimality and Duality

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problem, respect- ively. If $\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{b}^\top \boldsymbol{y}$, then $\boldsymbol{x}$ and $\boldsymbol{y}$ must be optimal solutions of the primal and dual, respectively.

**Optimality conditions for LP**. If $\boldsymbol{x}$, $\boldsymbol{y}$ satisfy:

1. $\boldsymbol{x}$ is primal feasible.
2. $\boldsymbol{y}$ is dual feasible.
3. The objective values coincide, i.e., $\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{b}^\top \boldsymbol{y}$.

Then, $\boldsymbol{x}$ and $\boldsymbol{y}$ are optimal solutions of the primal and dual problems respectively.

$\rightsquigarrow$ The reverse is also true (see the next theorem)!

## Strong Duality Theorem

If a linear program has an optimal solution, so does its dual and the optimal values of the primal and dual problems are equal.

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If a linear program has an optimal solution, so does its dual and the optimal values of the primal and dual problems are equal.

- We present a constructive proof: for a given primal optimal solution, we construct a dual optimal solution and show that their objective values are equal.
- We use the simplex method in our proof.
- In the proof, we will see that the simplex method actually also finds the dual optimal solution when it terminates.

# Proof

We prove by using the simplex method. Without loss of generality, we assume the primal problem is in the standard form.

If the primal problem has an optimal solution $\mathbf{x}^*$, then it must be associated with some optimal basis $B$ such that $\mathbf{x}_B = A_B^{-1}\mathbf{b}$ ($\mathbf{x}_B$ is the basic part of $\mathbf{x}^*$). Also, when the simplex method terminates, the reduced costs

$$\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A \geq 0 \tag{1}$$

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optimal dual solution

$$\mathbf{c}^T - \mathbf{c}_B^T A_B^{-1} A \geq 0 \tag{1}$$

$\mathbf{y}^T$

Now we define $\mathbf{y}^T = \mathbf{c}_B^T A_B^{-1}$. By above inequality, $A^T\mathbf{y} \leq \mathbf{c}$, i.e., $\mathbf{y}$ is feasible to the dual problem. In addition

$$\mathbf{b}^T\mathbf{y} = \mathbf{c}_B^T A_B^{-1}\mathbf{b} = \mathbf{c}_B^T\mathbf{x}_B = \mathbf{c}^T\mathbf{x}^*$$

$\mathbf{x}_B$

Therefore by the weak duality theorem, $\mathbf{y}$ must be optimal to the dual problem and the theorem holds. □

**Remark**

The dual optimal solution actually comes as a by-product when we use the simplex algorithm. The term

$$\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1}$$

will be the dual optimal solution (if the primal has an optimal solution). Thus, when solving the primal problem, the dual is solved automatically.

This is not a coincidence. Nearly all LP alg. (simplex method, interior point method, or ellipsoid method) solve both primal and dual problems simultaneously.

# Discussion: Continued

Based on the strong duality theorem, the pair $(\boldsymbol{x}, \boldsymbol{y})$ is optimal for the primal and dual, respectively, if and only if:

- $\boldsymbol{x}$ is primal feasible.
- $\boldsymbol{y}$ is dual feasible.
- They achieve the same objective values.

Solving LPs is equivalent to solving the following linear system:

- $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}$.
- $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$.
- $\boldsymbol{b}^\top \boldsymbol{y} = \boldsymbol{c}^\top \boldsymbol{x}$.

Later, we will show how this perspective helps when solving LPs.

# Question: Optimality & Feasibility & Boundedness

Which of the following states are possible for an LP and its dual?

by strong duality  by weak duality

| P \| D | Finite Optimum | Unbounded | Infeasible |
|---|---|---|---|
| Finite Optimum | ? ✓ | ? ✗ | ? ✗ |
| Unbounded | ? ✗ | ? ✗ | ? ✓ |
| Infeasible | ? ✗ | ? ✓ | ? |

strong / weak

To answer this question, we use the next structural property of LPs:

## LPs: States of Solutions

Consider an LP of minimizing $\boldsymbol{c}^\top \boldsymbol{x}$ over a nonempty polyhedron. Then, either the optimal cost is equal to $-\infty$ or there exists an optimal solution.

# Example: Primal and Dual are Infeasible

**Primal:**

$$\begin{aligned} \text{minimize} \quad & x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 = 1 \\ & 2x_1 + 2x_2 = 3 \end{aligned}$$

**Dual:**

$$\begin{aligned} \text{maximize} \quad & y_1 + 3y_2 \\ \text{subject to} \quad & y_1 + 2y_2 = 1 \\ & y_1 + 2y_2 = 2 \end{aligned}$$

# Answer

The only possible cases for LPs are:

| P \| D | Finite Optimum | Unbounded | Infeasible |
|---|---|---|---|
| Finite Optimum | ✓ | | |
| Unbounded | | | ✓ |
| Infeasible | | ✓ | ✓ |

- If both the primal and dual are feasible, then both problems must have optimal solutions. (This is one way to quickly determine if an LP is bounded).
- By strong duality, their optimal values must be the same.

We studied the relation between primal and dual optimal values.

Next, we study the relation between primal and dual optimal solutions.

# Complementarity Conditions

| Primal | | Dual | |
|---|---|---|---|
| min | $\boldsymbol{c}^\top \boldsymbol{x}$ | max | $\boldsymbol{b}^\top \boldsymbol{y}$ |
| s.t. | $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}$ | s.t. | $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$ |

## Theorem: Complementarity Conditions

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problem, respectively. Then, $\boldsymbol{x}$ and $\boldsymbol{y}$ are optimal solutions if and only if:

$$\begin{aligned} x_i > 0 \quad &\Longrightarrow \quad \boldsymbol{A}_i^\top \boldsymbol{y} = c_i \\ \boldsymbol{A}_i^\top \boldsymbol{y} < c_i \quad &\Longrightarrow \quad x_i = 0. \end{aligned}$$

Or in other words,

$$x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0, \quad \forall\ i.$$

# Proof

By the strong duality theorem, if $\mathbf{x}$ and $\mathbf{y}$ are optimal solutions for the primal and dual respectively, we must have

$$\mathbf{c}^T\mathbf{x} = \mathbf{b}^T\mathbf{y}$$

Thus we have

$$0 = \mathbf{c}^T\mathbf{x} - \mathbf{b}^T\mathbf{y} = \mathbf{c}^T\mathbf{x} - \mathbf{y}^T A\mathbf{x} = \sum_{i=1}^{n}(c_i - A_i^T\mathbf{y}) \cdot x_i \qquad (2)$$

# Proof

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Since $\mathbf{x}$ and $\mathbf{y}$ are both feasible, for each $i$, $c_i - A_i^T\mathbf{y} \geq 0$ and $x_i \geq 0$. Therefore, in order for (2) to hold, we must have

$$(c_i - A_i^T\mathbf{y}) \cdot x_i = 0, \quad \forall i$$

The other direction follows from the same arguments $\square$.

# Example

**Primal:**

$$\begin{array}{rrrrrrr} \text{minimize} & 13x_1 & +10x_2 & +6x_3 & & \\ \text{s.t.} & 5x_1 & +x_2 & +3x_3 & = & 8 \\ & 3x_1 & +x_2 & & = & 3 \\ & x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

$\rightsquigarrow$ Optimal solution: $(1, 0, 1)$.

**Dual problem:**

$$\begin{array}{rrrrr} \text{maximize} & 8y_1 & +3y_2 & & \\ \text{s.t.} & 5y_1 & +3y_2 & \leq & 13 \\ & y_1 & +y_2 & \leq & 10 \\ & 3y_1 & & \leq & 6 \end{array}$$

$\rightsquigarrow$ Optimal solution: $(2, 1)$.

Verify the complementarity conditions:

$$x_1 \cdot (13 - 5y_1 - 3y_2) = 0,\ x_2 \cdot (10 - y_1 - y_2) = 0,\ x_3 \cdot (6 - 3y_1) = 0.$$

Today's lecture

- Weak and Strong Duality
- Complementarity Condition

Next lecture

- Duality via Simplex Tableau
- Intepretation of Duality

Questions?